AMAZON INDIA SALES ANALYSIS

Executive Dashboard in Power BI

Project Explanation Document

Tools: Python (pandas), Power BI Desktop, DAX

Dataset: Amazon India product catalogue (amazon.csv)

1. Project Overview

This project turns a raw Amazon India product dataset into an interactive executive dashboard in Power BI. The dashboard lets a business user explore which product categories, brands and price ranges drive estimated revenue, how deeply products are discounted, and how customers rate them. Every visual responds to slicers for category, price range and rating, so the same page answers many different questions.

The layout was inspired by a banking executive dashboard (KPI cards on top, charts in a grid, and a key insights panel), adapted to e-commerce data.

2. Objectives

- Clean and prepare a messy real-world product dataset for analysis.
- Define meaningful KPIs and write them as reusable DAX measures.
- Build a one-page interactive dashboard with consistent design and a clear story.
- Add dynamic insights and conditional formatting that react to slicer selections.

3. Dataset Description

The raw file contains 1,465 rows and 16 columns, one row per product listing. Important fields are listed below.

Column	Meaning
product_id, product_name	Unique product identifier and listing title
category	Pipe-separated category path, for example Electronics|Mobiles|Smartphones
actual_price, discounted_price	List price and selling price in rupees (stored as text with the ₹ symbol)
discount_percentage	Discount shown on the listing (stored as text with the % symbol)
rating, rating_count	Average customer rating and number of ratings
review_id, user_id, review_content	Review data packed into comma-separated cells

The dataset has no date or region column, so time trends and geographic analysis are outside its scope.

4. Data Cleaning and Preparation

Cleaning was done in Python with pandas, and the result was exported as amazon_clean.csv for Power BI.

- **Duplicates:** The file had 114 repeated product_id values (the same product listed under more than one category). Only the first occurrence was kept, leaving 1,351 unique products.
- **Text to numbers:** Prices such as ₹1,099 were converted to numbers, discount strings such as 64% to 64, and rating counts such as 24,269 to 24269.
- **Missing and bad values:** One row had an invalid rating value; it was filled with the median rating. Two products with a missing rating count were set to 0.
- **Category split:** The category path was split into main, sub and leaf category, and the brand was extracted from the product name.
- **New columns:** rupees_saved, est_revenue, and price, discount and rating bands were added for analysis.

Estimated revenue is calculated as discounted price × rating count. It is a proxy for sales volume, since actual sales numbers are not available in the data.

5. Power BI Implementation

5.1 Data model

A single table named amazon was loaded from the cleaned CSV. Band columns (price, rating, discount) were sorted logically using helper order columns built from the underlying numeric fields, so slicers and charts show Under 500 to Above 20K instead of alphabetical order.

5.2 Key DAX measures

Measure	Purpose
Total Products	Distinct count of product_id
Avg Rating / Avg Discount %	Average of rating and discount_pct
Median Price	Median of discounted_price
Est Revenue	SUMX of discounted_price × rating_count
Total Reviews / Total Savings	Sum of rating_count; savings weighted by rating_count
Top Rated %	Share of products with rating 4.5 or higher
Insight Top Category	Text measure naming the top sub-category and its revenue share
Insight Discount	Text measure showing the share of products with 50%+ discount
Colour measures	Return hex codes that drive card background colours

5.3 Dashboard design

- Eight KPI cards: Total Products, Total Reviews, Avg Rating, Avg Discount, Est. Revenue, Total Savings, Median Price and Top Rated %.
- Two dynamic insight cards that rewrite themselves when slicers change.
- Three slicers: Category, Price Range and Rating.
- Six visuals: revenue by sub-category, top 10 products by revenue, rating distribution, price band mix (donut), top brands table, and average rating by discount band.
- A consistent orange and navy theme, white rounded cards and one font family (Segoe UI).

5.4 Conditional formatting

Card backgrounds change automatically with slicer selections. For average-type cards, the current value is compared with the overall value using ALL(amazon) and shown in green or red. For total-type cards, the share of overall data remaining decides the colour. This makes it visible at a glance when a selection performs above or below the overall catalogue.

6. Key Findings

Metric	Result
Unique products analysed	1,351
Average rating	4.09 out of 5
Average discount	46.69%
Median selling price	₹899
Total ratings in dataset	24 million
Estimated revenue (proxy)	about ₹69 billion

- Mobiles & Accessories alone contributes about 45% of estimated revenue, followed by Home Theater, TV & Video.
- Electronics, Computers & Accessories and Home & Kitchen make up about 99% of all products.
- Around 49% of products carry a discount of 50% or more, and Wearable Technology has the deepest average discount (about 70%).
- Redmi leads brands by estimated revenue (about ₹18.9 billion from 25 listings), ahead of Samsung and MI.
- Most products (about 60%) are rated between 3.5 and 4.5, and only about 7% reach 4.5 or above.
- Products with heavier discounts tend to have slightly lower average ratings.

7. Limitations

- Estimated revenue is a proxy (price × rating count), not actual sales.
- Product variants can share the same rating count, which may overstate revenue for some brands.

- The dataset has no dates or regions, so seasonality and geographic views are not possible.
- Review text was not analysed because it is stored in combined comma-separated cells.

8. Future Scope

- Add a second page for pricing and discounts with a scatter plot of discount against rating.
- Add a brand performance page and page navigation buttons.
- Parse the review data and run sentiment analysis on review text.
- Publish the report to Power BI Service for sharing and scheduled refresh.

9. Skills Demonstrated

- Data cleaning and feature engineering with Python and pandas.
- DAX measures, calculated columns and sort-by-column logic.
- Dashboard design: KPI cards, slicers, layout, theming and consistency.
- Interactive storytelling with dynamic text and conditional formatting.

10. Resume Summary

Built an interactive Power BI executive dashboard on 1,351 Amazon India products. Cleaned the raw data with Python, wrote DAX measures for 8 KPIs, and added dynamic insights and conditional formatting driven by slicers. Findings showed Mobiles & Accessories contributes about 45% of estimated revenue and that nearly half of all products are discounted by 50% or more.